

TrES-3b

R-2472

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110531 · created 2026-08-23T23:26:55+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455712.84127 +/- 0.00056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.011
Transit depth (Rp/Rs)^2	2.75 +/- 0.35 [%]
Orbital Inclination (inc)	82.84 +/- 0.45
Airmass coefficient 1 (a1)	1.071 +/- 0.0092
Airmass coefficient 2 (a2)	-0.067 +/- 0.033
Scatter in the residuals of the lightcurve fit is	0.85 %
Best Comparison Star	#1 - [369, 377]
Optimal Aperture	4.13
Optimal Annulus	11.71
Transit Duration (day)	0.0632 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.011 vs 0.16309 (deviation 0.21σ)
Duration fit vs published	0.0632 d vs 0.0567 d (deviation 1.59σ)
Mid-time O–C	-3.0 min
Depth SNR	12.1
Residual scatter	0.85 %

Light curve

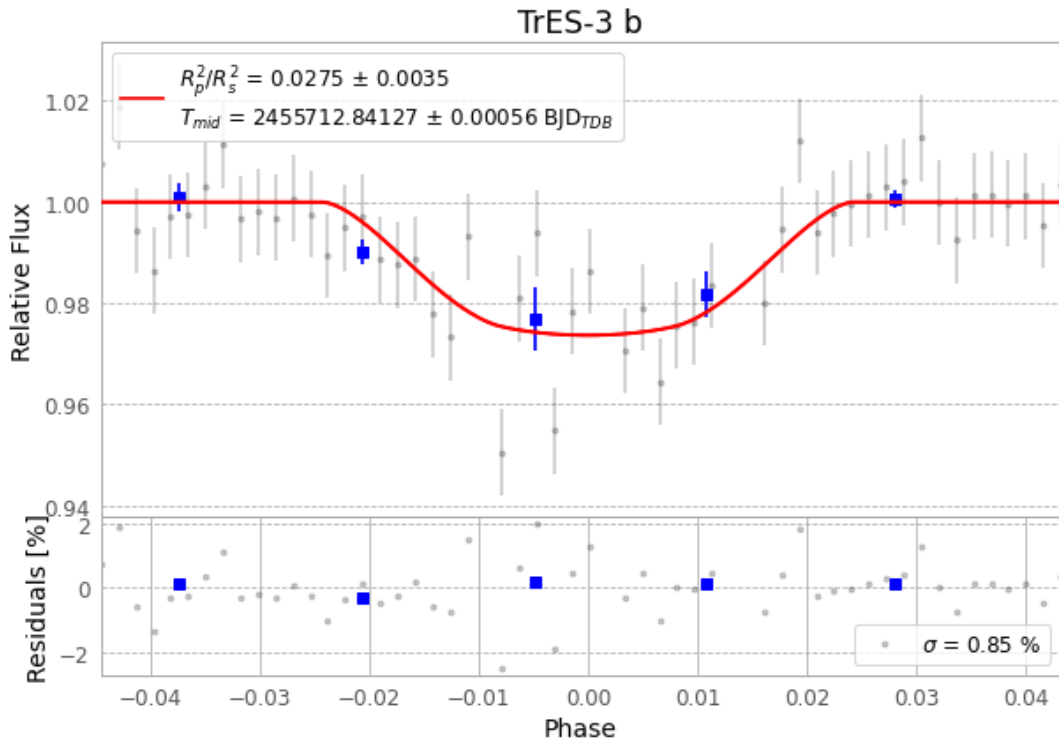


Figure 1 — FinalLightCurve_TrES-3 b_2011-05-30.png (SHA-256 a39aa41457ed061c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [266, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 39e69fb7dc603445...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

56 FITS frames (37 MB), TRES-3110531064213.FITS → TRES-3110531092713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110531.log (87a215c9eef3...), FinalLightCurve_TrES-3 b_2011-05-30.png (a39aa41457ed...), FinalLightCurve_TrES-3 b_2011-05-30.csv (a09921b05c0d...)

Compute resources

Wall time 169.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 23:26Z.